

Extracellular matrix viscoelasticity regulates spatiotemporal mammary branching morphogenesis

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The effect of viscoelasticity on tissue organization is unknown

Branching morphogenesis is a fundamental process that occurs during the development of mammary tissue, and several signaling pathways that regulate mammary branching are disrupted during the progression of breast cancer. Mammary gland development is influenced by mechanical and chemical cues within its *in vivo* microenvironment, and conventionally, tissue development has been studied with *in vivo* or *ex vivo/in vitro* systems where mechanical properties are poorly controlled in space and time, lack tunability, or do not mimic the native extracellular matrix (ECM) mechanics. Here, we engineered a collagen-alginate interpenetrating network (IPN) to recapitulate the mammary *in vivo* microenvironment and determine how ECM viscoelasticity regulates tissue branching in mammary development.

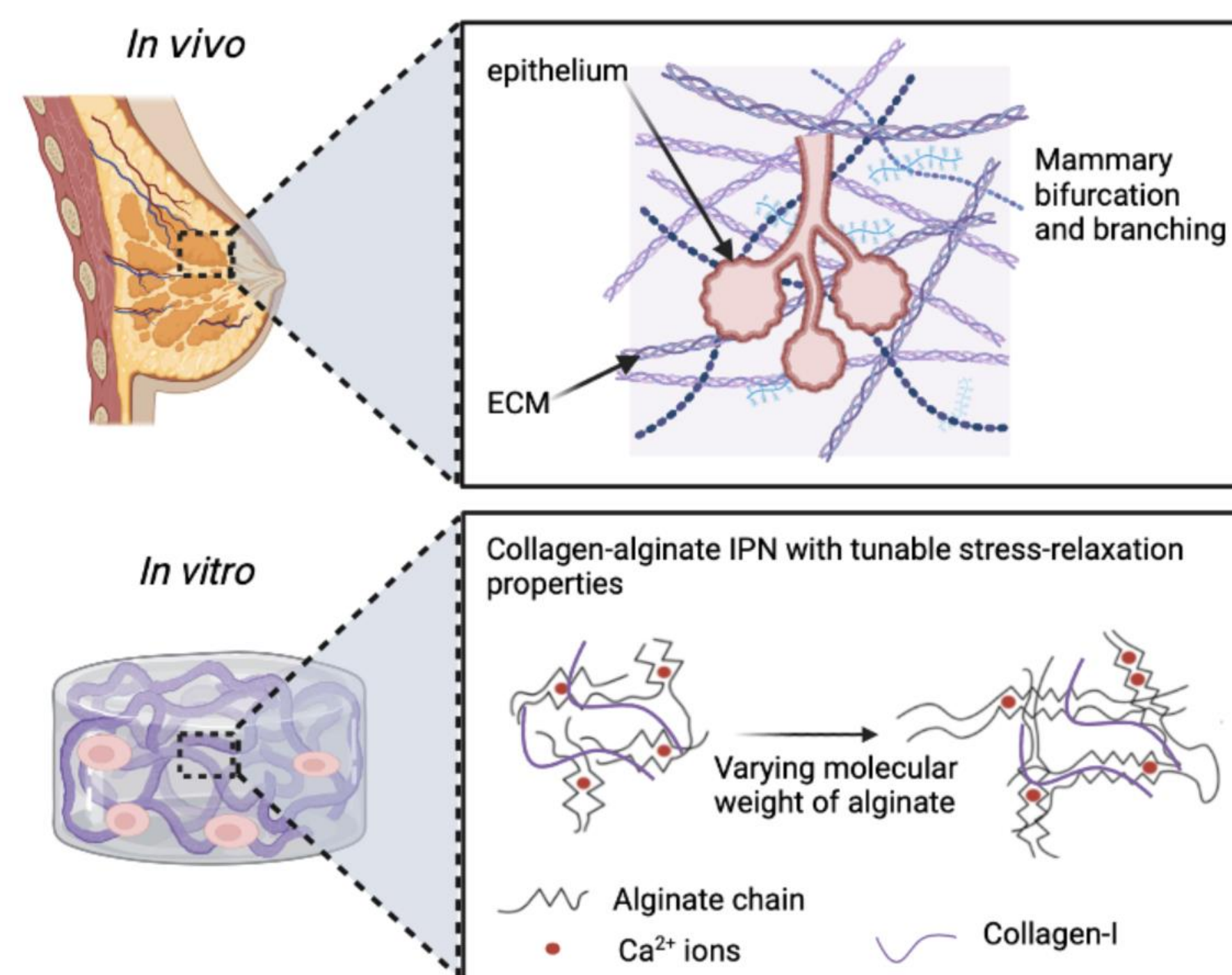


Figure 1: Schematic depicting how mammary gland development is influenced by mechanical and chemical cues within its *in vivo* microenvironment, and 3D collagen-alginate interpenetrating networks (IPNs) enable independent tuning of the viscoelasticity of the matrix. Figure created in *BioRender*.

3D collagen-alginate IPNs enable independent tuning of the viscoelasticity of the matrix

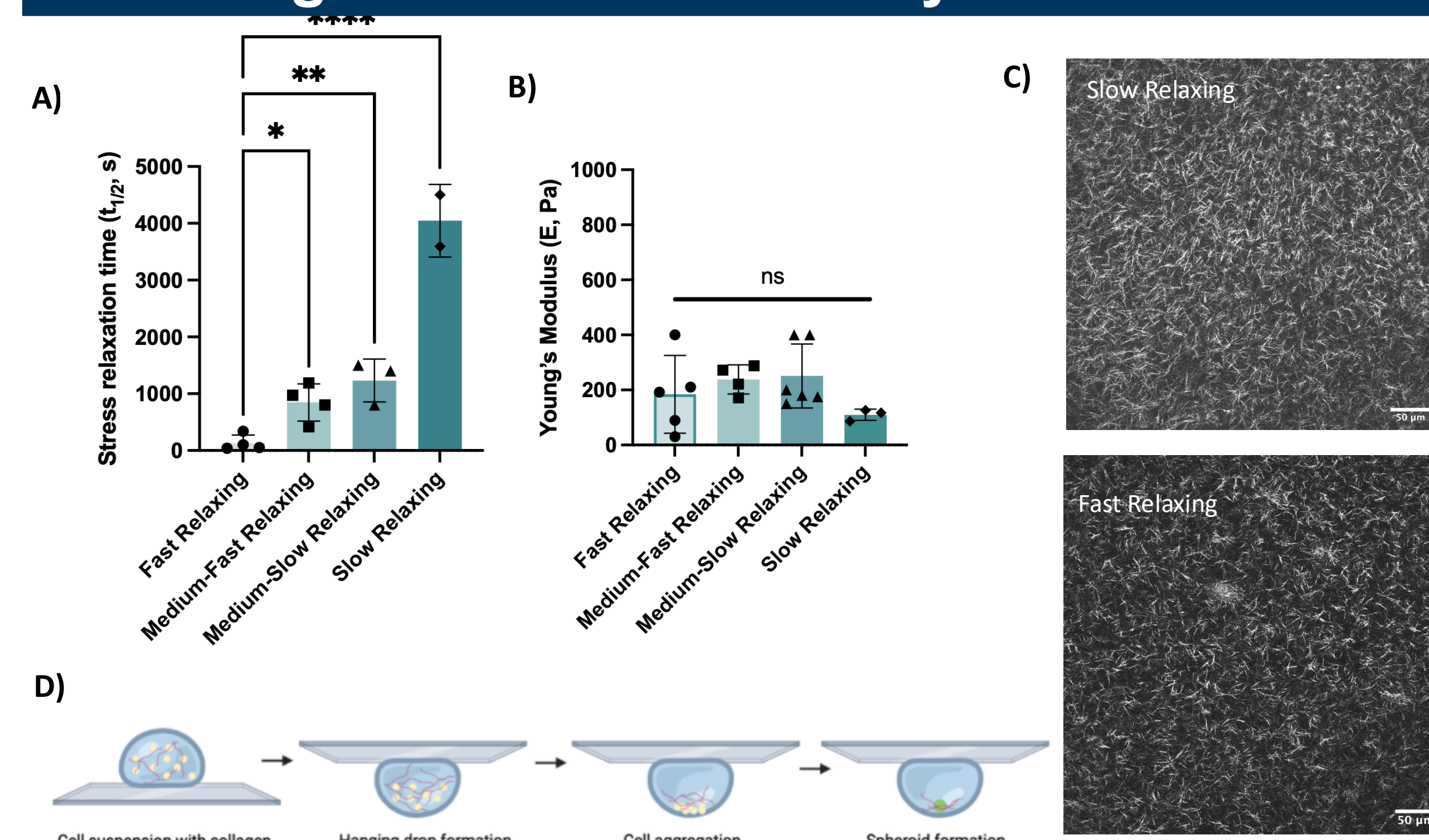


Figure 2: (a) Molecular weight of alginate can be tuned to fabricate matrices with varying stress-relaxation times. (b) Rheology of hydrogels reveals that hydrogels with varying stress-relaxation times have matrix stiffnesses that are not significantly different. (c) Confocal reflectance microscopy reveals that slow relaxing matrices and fast relaxing matrices have similar collagen microarchitectures. (d) MCF10A spheroids were prepared via the hanging-drop method and encapsulated in matrices after 48 hours. Image created in *BioRender*.

Mammary epithelial spheroid morphology varies with stress-relaxation rate

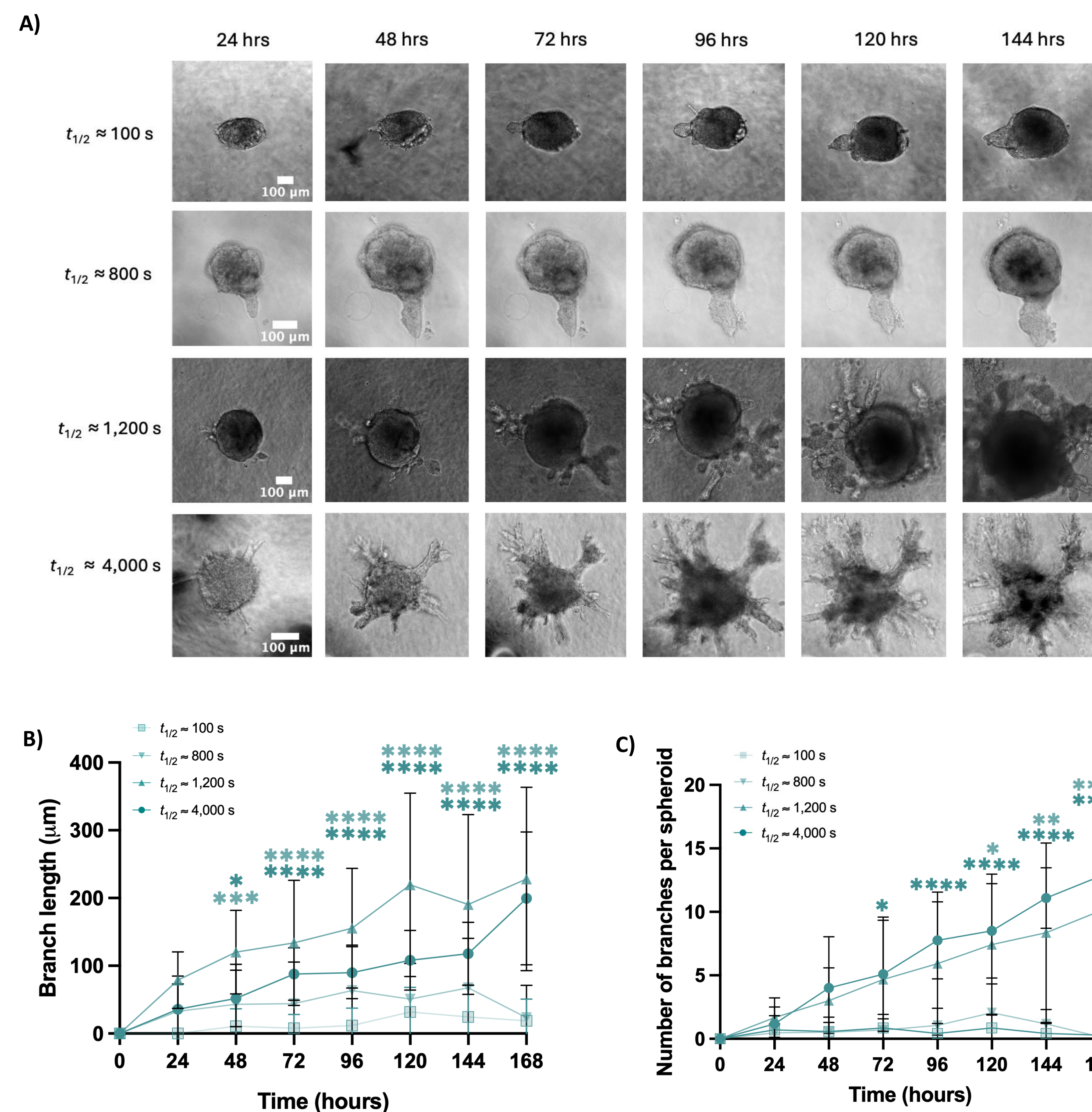


Figure 3: (a) Brightfield images of MCF10A spheroids in 3D slow stress-relaxing matrices ($t_{1/2} = 1200$ s, $t_{1/2} = 4000$ s) and fast stress-relaxing matrices ($t_{1/2} = 100$ s, $t_{1/2} = 800$ s) over 7 days. (b) Quantification of primary branch length demonstrates a faster branch growth rate in slow stress-relaxing matrices. (c) Quantification of the number of branches demonstrates more branches are formed in slow stress-relaxing matrices.

pFAK and $\beta 1$ integrin localize to the tip of mammary branches in slow relaxing gels

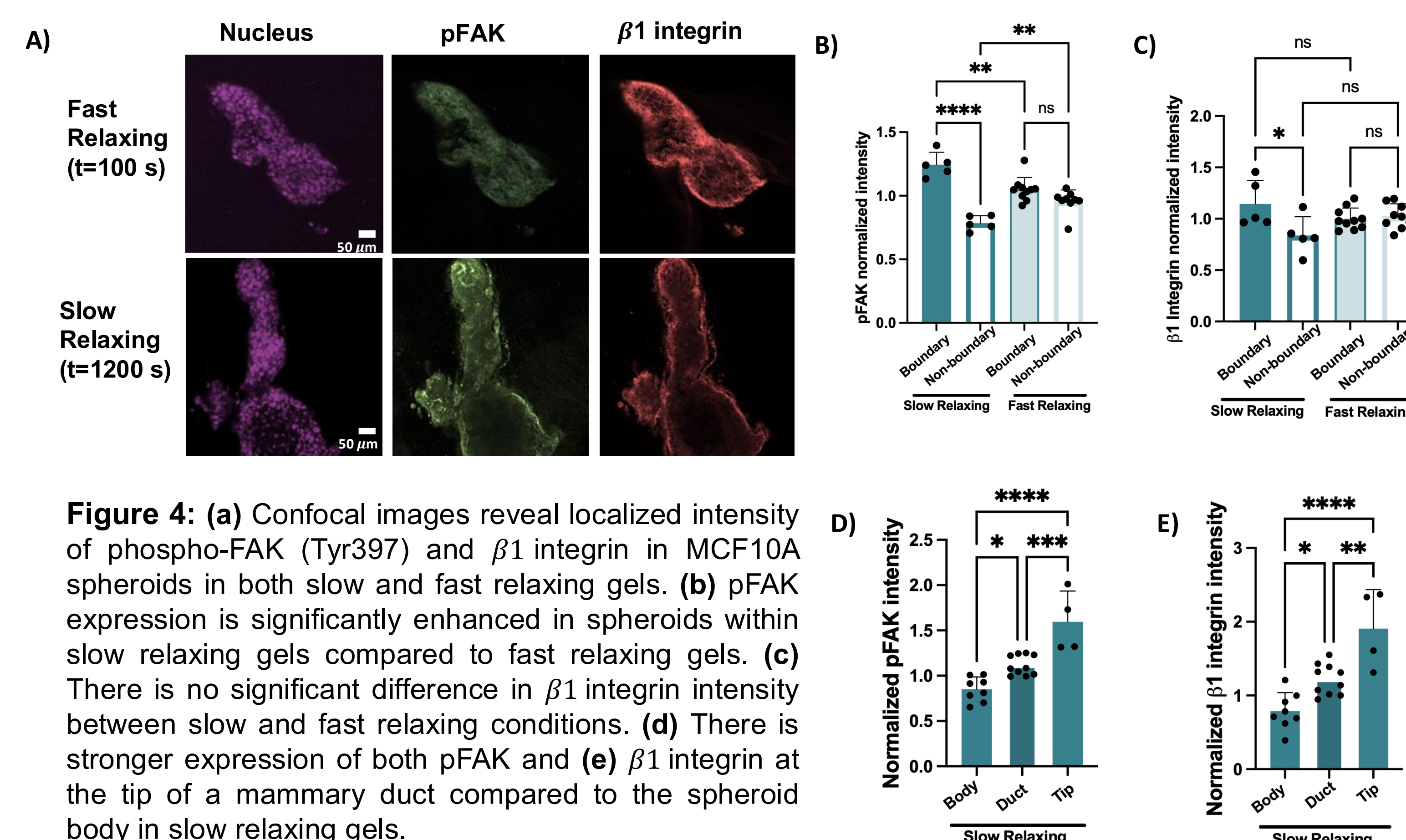


Figure 4: (a) Confocal images reveal localized intensity of phospho-FAK (Tyr397) and $\beta 1$ integrin in MCF10A spheroids in both slow and fast-relaxing gels. (b) pFAK expression is significantly enhanced in spheroids within slow relaxing gels compared to fast relaxing gels. (c) There is no significant difference in $\beta 1$ integrin intensity between slow and fast relaxing conditions. (d) There is stronger expression of both pFAK and (e) $\beta 1$ integrin at the tip of a mammary duct compared to the spheroid body in slow relaxing gels.

Slow stress-relaxing matrices promote collagen fiber alignment along branching axis

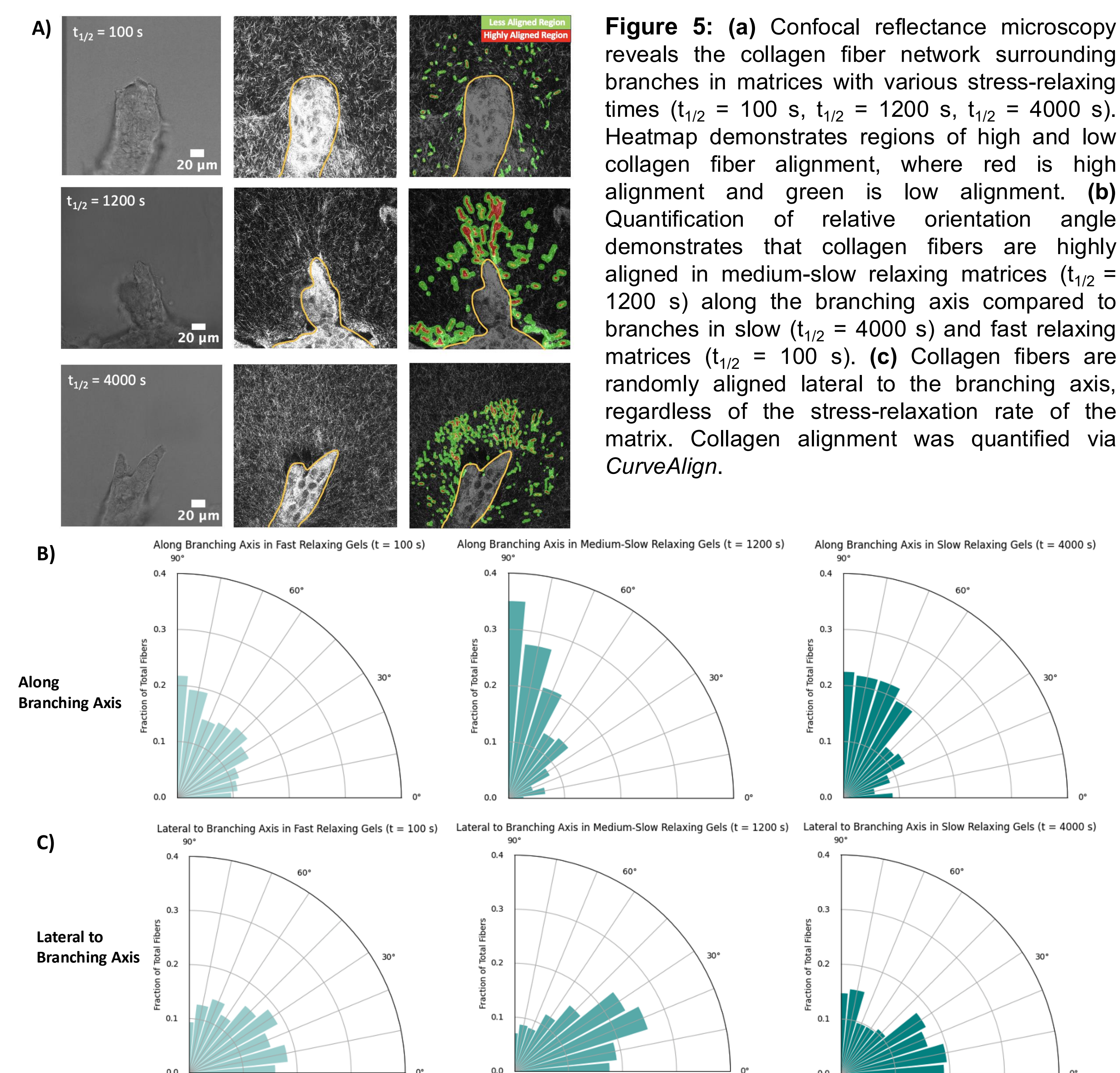


Figure 5: (a) Confocal reflectance microscopy reveals the collagen fiber network surrounding branches in matrices with various stress-relaxing times ($t_{1/2} = 100$ s, $t_{1/2} = 1200$ s, $t_{1/2} = 4000$ s). Heatmap demonstrates regions of high and low collagen fiber alignment, where red is high alignment and green is low alignment. (b) Quantification of relative orientation angle demonstrates that collagen fibers are highly aligned in medium-slow relaxing matrices ($t_{1/2} = 1200$ s) along the branching axis compared to branches in slow ($t_{1/2} = 4000$ s) and fast relaxing matrices ($t_{1/2} = 100$ s). (c) Collagen fibers are randomly aligned lateral to the branching axis, regardless of the stress-relaxation rate of the matrix. Collagen alignment was quantified via *CurveAlign*.

Osmotic pressure impedes mammary branching in a dose-dependent manner

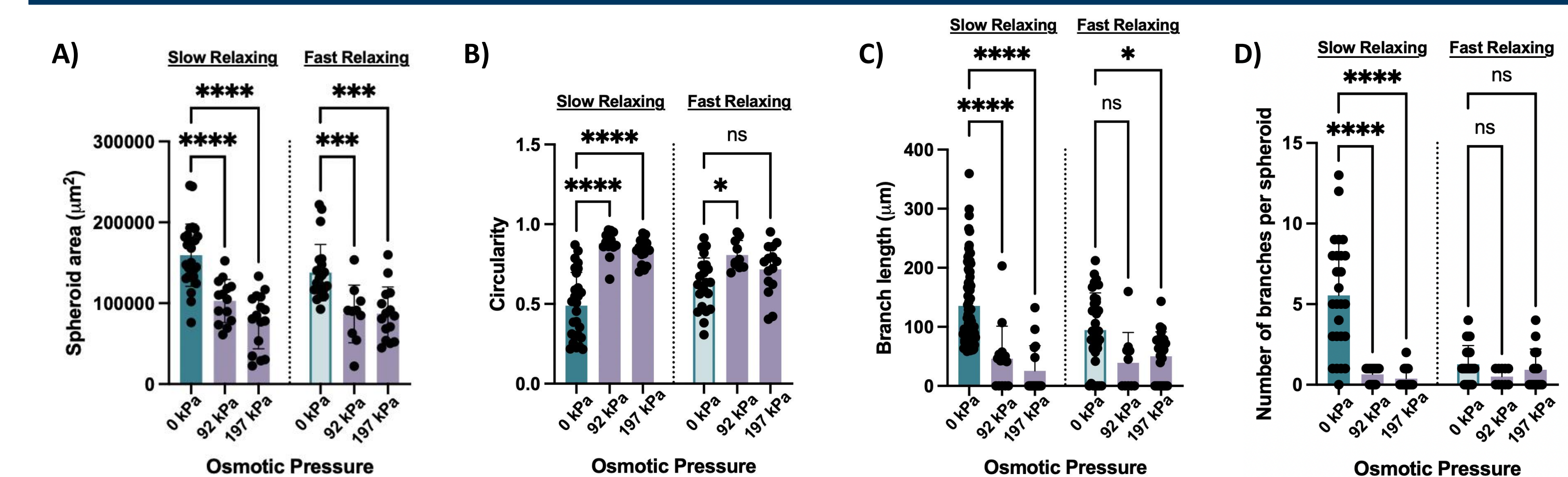


Figure 6: (a) Quantification of MCF10A spheroid area under varied osmotic pressure for 7 days. (b) Circularity is enhanced in slow relaxing gels when spheroids are subjected to hypertonic media. (c) Quantification of branch length demonstrates that spheroids branch to a lesser extent in slow relaxing gels under an osmotic pressure of 92 and 197 kPa. (d) In slow relaxing gels, spheroids branch less when subjected to osmotic pressure, but there is no significant difference in fast relaxing gels. Hyperosmotic pressures were modulated by adding 400 Da polyethylene glycol, and osmotic pressures were calculated via an empirically derived formula.

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